PORTFOLIO PROJECT · REAL DATA: PUBLIC HEALTH AGENCY OF CANADA, 2016 TO 2026

Opioid and stimulant harms in Canada: what has changed since 2016?

Start here: the headline numbers, what they mean, and what I would do about them.

The question: Which harm measures are falling, which are not, and where is the burden greatest, so a public health team knows where to plan?

Reset filters

Opioid deaths, latest full year

Full year 2025. Peak was 8,083 in 2023

5,704

Deaths against the peak

Hospital stays -21.75%, ED visits -28.51%

-29.43%

EMS against the peak

2026 Q1: +60.93% vs a year ago

-7.84%

Hospital stays per death

It was 1.62 in 2016

0.88

What the data shows

- Opioid deaths are 29.43% below the 2023 peak (8,083 to 5,704), but EMS responses are only 7.84% below theirs. The measures do not fall together.
- Rates vary widely: in 2024 British Columbia had 35.7 deaths per 100,000, Ontario 14.0 and Quebec 3.4.
- Around 71.64% of deaths are men, and 52.58% are aged 30 to 49.

What I would do

- Track EMS responses and emergency department visits next to deaths: deaths alone understate the ongoing burden.
- Start with the high-rate places (British Columbia, Yukon, Manitoba, Alberta, Saskatchewan: 22 to 36 per 100,000 in 2024), using the same year for every place.
- Aim outreach at men aged 30 to 49, the largest group. These are proposals for discussion, not clinical advice.

How to use this report

- Use the page tabs. Four measures compares the sources; Trend, Provinces and Who go deeper.
- Data quality shows who reports what and what I did about missing values.
- With nothing chosen, pages show deaths and opioids, and Provinces and Who show 2024. Reset filters clears the slicers.

About the data and model

- Source: Public Health Agency of Canada, Opioid- and Stimulant-related Harms in Canada (data updated September 2026). Counts and rates only, no individual records.
- The file is a flattened export of website charts; I rebuilt it as three fact tables and six dimensions.
- 2026 covers January to March only, so full years end at 2025.

Footnote: a Canada total only adds up the places that report. Hospital stays leave out Quebec, and emergency department visits cover 7 of 13 provinces and territories. A suppressed or missing value is never shown as zero.

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Deaths have fallen faster than ambulance calls

Opioid deaths, hospital stays, emergency department visits and EMS responses for Canada, side by side.

Region (blank = Canada)

All

Since the 2023 peak: deaths -29.43%, hospital stays -21.75%, EMS responses -7.84%.

Reset filters

Deaths vs a year earlier
2026 Q1
-5.31%

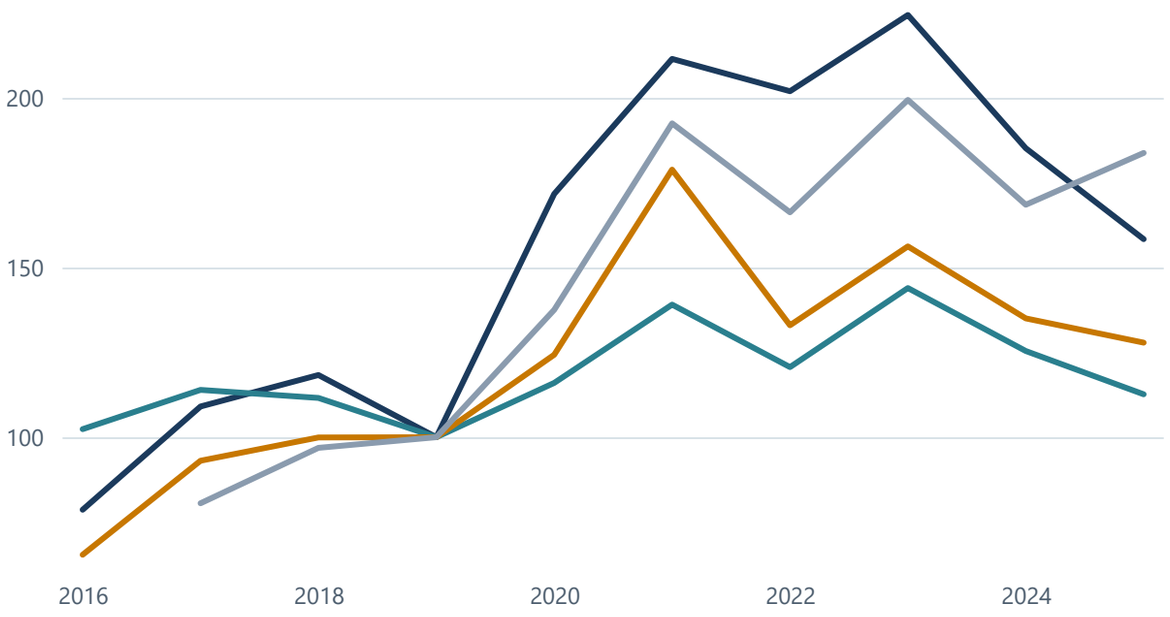
Hospital stays vs a year earlier
2026 Q1
+13.55%

ED visits vs a year earlier
2026 Q1
+43.51%

EMS responses vs a year earlier
2026 Q1
+60.93%

Opioid counts, indexed to 2019

Each line starts at 100 in 2019. Full years only.



Opioid counts by year

ED visits and hospital stays cover only the places that report each year.

Year	Deaths	Hospital stays	ED visits	EMS responses	Stays per death
2016	2,832	4,575	11,912		1.62
2017	3,928	5,092	16,961	17,269	1.30
2018	4,262	4,986	18,210	20,771	1.17
2019	3,602	4,469	18,224	21,444	1.24
2020	6,184	5,184	22,652	29,493	0.84
2021	7,617	6,215	32,592	41,275	0.82
2022	7,274	5,392	24,234	35,653	0.74
2023	8,083	6,433	28,470	42,756	0.80
2024	6,669	5,604	24,604	36,131	0.84
2025	5,704	5,034	23,301	39,406	0.88
2026 (Jan to Mar)	1,354	1,374	7,701	12,535	1.01

01 · Against the peak

-29.43%

Opioid deaths are 29.43% below their 2023 peak, hospital stays 21.75% and ED visits 28.51% below theirs, but EMS responses only 7.84%.

02 · The latest quarter

+60.93%

In 2026 Q1 EMS responses were 60.93% above a year earlier and hospital stays 13.55% above, while deaths were 5.31% below.

03 · Read with care

Direction

The four sources count different things and cover different places, so compare their direction, not their size.

Footnote: the sources count a death, a hospital stay, a visit and an ambulance call. The EMS rise in 2026 comes mostly from British Columbia and Winnipeg and may reflect how services record calls.

Data: PHAC, Opioid- and Stimulant-related Harms in Canada (Open Government Licence - Canada). No individual records.

Counts to 2026 Q1 | Latest full year 2025 | 2026 is January to March only

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Counts rose sharply in 2020 and have eased since the 2023 peak

Quarterly counts for one source and one substance, with the yearly totals underneath.

Source (none = Deaths): All

Substance (none = Opioids): All

Showing: Deaths, opioids. 2026 Q1 is -5.31% vs a year earlier.

Reset filters

Latest quarter
2026 Q1
1,354

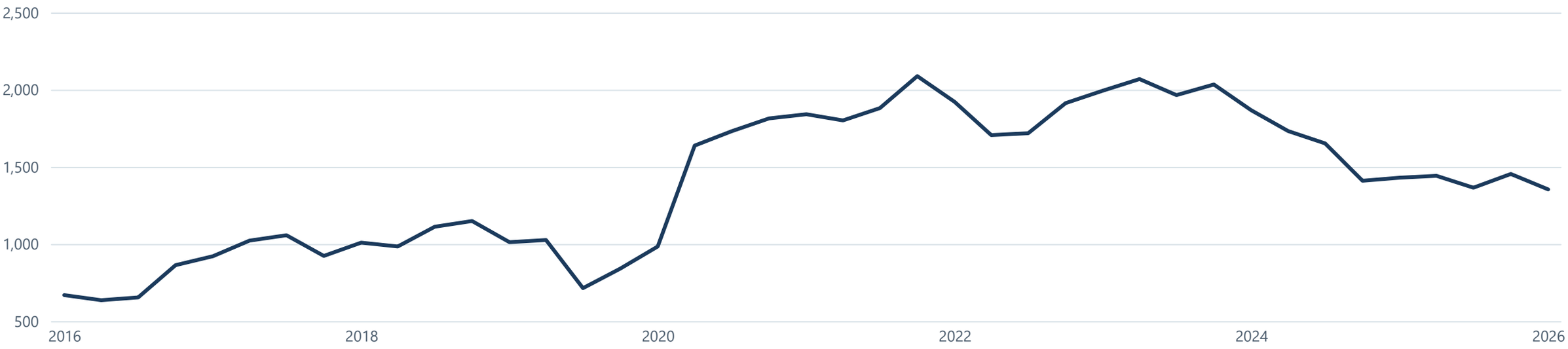
Against a year earlier
-5.31%

Highest quarter
Highest quarter was Oct 2021
2,088

Latest full year
Full year 2025
5,704

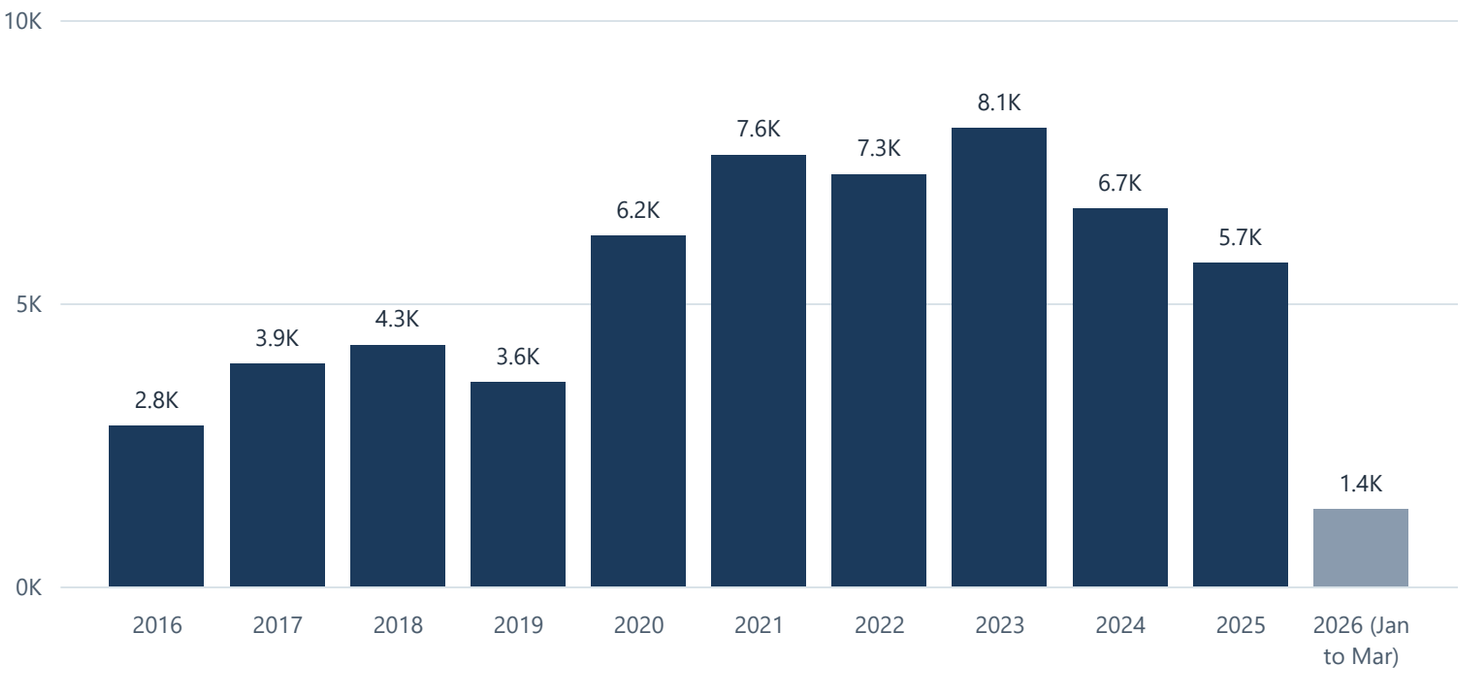
Quarterly count, Canada

Canada here means the places that report for this source. Pick a source and substance above.



Yearly count

2026 covers January to March only and is shown in grey.



Yearly count and rate

EMS has no published rate, so that column stays blank for EMS responses.

Year	Count	Rate per 100,000
2016	2,832	7.8
2017	3,928	10.7
2018	4,262	11.5
2019	3,602	9.6
2020	6,184	16.3
2021	7,617	19.9
2022	7,274	18.7
2023	8,083	20.2
2024	6,669	16.2
2025	5,704	13.7
2026 (Jan to Mar)	1,354	13.0

01 · The 2020 jump

+71.68%

Opioid deaths rose from 3,602 in 2019 to 6,184 in 2020, the sharpest rise in the series (deaths and opioids chosen).

02 · The peak

8,083

Deaths peaked at 8,083 in 2023 and were 5,704 in 2025, 29.43% below the peak.

03 · Read with care

Q1 only

2026 covers January to March only, so full years end at 2025 and 2026 is shown in grey.

Footnote: counts are for the source and substance chosen above. Canada means only the places that report, and a missing value is never shown as zero.

Data: PHAC, Opioid- and Stimulant-related Harms in Canada (Open Government Licence - Canada). No individual records.

Counts to 2026 Q1 | Latest full year 2025 | 2026 is January to March only

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Rates differ several times over between provinces

The rate per 100,000 for one year, with suppressed and missing values labelled, never shown as zero.

Source (none = Deaths): All

Substance (none = Opioids): All

Year (none = 2024): All

Showing: Deaths, opioids, 2024

Reset filters

Canada rate per 100,000
Only the places that report
16.2

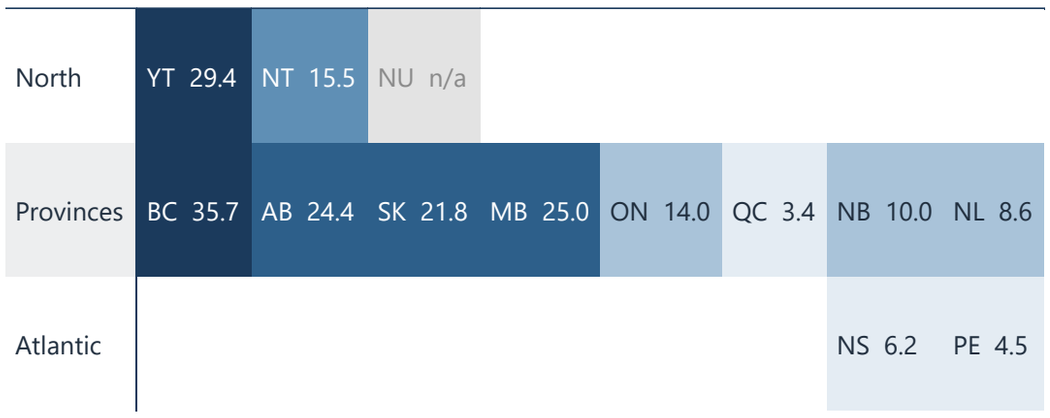
Highest rate
British Columbia
35.7

Lowest rate
Quebec
3.4

Regions with a rate
of the provinces and territories
12

Rate per 100,000: tile map of Canada

Crude rate per 100,000 population, not age-adjusted.



Lighter = lower rate darker = higher rate n/a = not reported

Tiles sit roughly where each province is. Colours compare tiles on this page.

Canada: 16.2 per 100,000, counting only the places that report.

Highest: British Columbia (35.7). Lowest: Quebec (3.4).

Count, rate and status by region

Status shows Suppressed or Not available. Regions with no data are left out.

Region	Count	Rate per 100,000	Status
British Columbia	2,025	35.7	Reported
Yukon	14	29.4	Reported
Manitoba	373	25.0	Reported
Alberta	1,197	24.4	Reported
Saskatchewan	272	21.8	Reported
Canada	6,669	16.2	Reported
Northwest Territories	7	15.5	Reported
Ontario	2,267	14.0	Reported
New Brunswick	86	10.0	Reported
Newfoundland and Labrador	47	8.6	Reported
Nova Scotia	67	6.2	Reported
Prince Edward Island	8	4.5	Reported
Quebec	306	3.4	Reported
Nunavut			Not available

01 · The highest rates

35.7

In 2024 British Columbia had 35.7 deaths per 100,000, then Yukon 29.4, Manitoba 25.0 and Alberta 24.4.

02 · The lowest rates

3.4

Quebec had 3.4 and Ontario 14.0 per 100,000 in 2024: British Columbia's rate is more than ten times Quebec's.

03 · Read with care

2024 first

Some 2025 values look provisional (Ontario deaths -37.49%, Quebec +93.14%), so the page opens on 2024.

Footnote: rates are crude, not age-adjusted, and reporting differs by province, so use the ranking to ask questions, not to judge. The Canada rate only adds up the places that report.

Data: PHAC, Opioid- and Stimulant-related Harms in Canada (Open Government Licence - Canada). No individual records.

Counts to 2026 Q1 | Latest full year 2025 | 2026 is January to March only

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Seven in ten deaths are men, and half are aged 30 to 49

Counts by age group and sex for one source, substance and year.

Source (none = Deaths): All

Substance (none = Opioids): All

Year (none = 2024): All

Region (blank = Canada): All

Reset filters

Male share
Showing: Deaths, opioids, 2024
71.64%

Aged 30 to 49
Showing: Deaths, opioids, 2024
52.58%

With age and sex shown
98.73%

Deaths or cases shown
Showing: Deaths, opioids, 2024
6,584

Heat map: count by age group and year

Darker means more. All years for the chosen source and substance. 2026 Q1 is January to March only.

Age group	2018	2019	2020	2021	2022	2023	2024	2025	2026 Q1
0 to 19 years	66	61	108	123	132	156	98	86	15
20 to 29 years	807	668	1,110	1,312	1,119	1,249	981	672	139
30 to 39 years	1,115	991	1,635	2,133	1,946	2,302	1,854	1,349	262
40 to 49 years	878	729	1,385	1,667	1,719	1,914	1,608	1,303	275
50 to 59 years	880	772	1,303	1,553	1,486	1,518	1,196	1,006	223
60 years or more	375	339	606	759	795	882	847	780	119

Count and share by age group and sex

Blank means suppressed or not available, not zero.

Age group	Sex	Count	Share of total
0 to 19 years	Female	44	0.67%
0 to 19 years	Male	54	0.82%
20 to 29 years	Female	332	5.04%
20 to 29 years	Male	649	9.86%
30 to 39 years	Female	520	7.90%
30 to 39 years	Male	1,334	20.26%
40 to 49 years	Female	441	6.70%
40 to 49 years	Male	1,167	17.72%
50 to 59 years	Female	311	4.72%
50 to 59 years	Male	885	13.44%
60 years or more	Female	219	3.33%
60 years or more	Male	628	9.54%

What it shows: every age group roughly doubled from 2019 to 2021, ages 30 to 39 are highest in every year, and counts have fallen since 2023.

01 · Men
71.64%
In 2024, 71.64% of opioid deaths were men (deaths and opioids chosen).

02 · Aged 30 to 49
52.58%
52.58% of opioid deaths in 2024 were aged 30 to 49, the largest group.

03 · Read with care
98.73%
The age and sex split covers 98.73% of 2024 deaths and 91.09% of 2025. The rest is suppressed.

Footnote: PHAC suppresses small counts, so age and sex cells add up to less than the total. Smaller regions show many blank cells. Sex is reported as male or female only.

Data: PHAC, Opioid- and Stimulant-related Harms in Canada (Open Government Licence - Canada). No individual records.

Counts to 2026 Q1 | Latest full year 2025 | 2026 is January to March only

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Before any analysis: what each number in the file means

Who reports what, how much is missing, and what I did about it.

Who reports what: opioid counts, 2024

Teal = reported, amber = suppressed (small count), grey = not available, white = not in this source.

Region	Deaths	Hospital stays	Emergency department visits	EMS responses
Canada	Reported	Reported	Reported	Reported
British Columbia	Reported	Reported	Reported	Reported
Alberta	Reported	Reported	Reported	Reported
Saskatchewan	Reported	Reported	Reported	Reported
Manitoba	Reported	Reported	Not available	Not in file
Ontario	Reported	Reported	Reported	Reported
Quebec	Reported	Not available	Not available	Not in file
New Brunswick	Reported	Reported	Not available	Reported
Nova Scotia	Reported	Reported	Reported	Reported
Prince Edward Island	Reported	Reported	Reported	Not in file
Newfoundland and Labrador	Reported	Reported	Not available	Not available
Yukon	Reported	Not in file	Reported	Not in file
Northwest Territories	Reported	Not in file	Not available	Not in file
Nunavut	Not available	Not in file	Not available	Not in file
Territories	Not in file	Reported	Not in file	Not in file
Northern and rural Manitoba	Not in file	Not in file	Not in file	Not available
Winnipeg, Manitoba	Not in file	Not in file	Not in file	Reported
Whitehorse, Yukon	Not in file	Not in file	Not in file	Reported
Yellowknife, Northwest Territories	Not in file	Not in file	Not in file	Reported

What I found in the file and what I did

- 34,890 rows and 39 columns, with English and French copies of every label. It is a chart export, not a table, so I rebuilt it as three tables (yearly, quarterly, age and sex).
- 62.08% of values are numbers (5.11% are real zeros), 24.82% are blank (not available) and 13.09% are suppressed. I keep each kind separate and never show missing as zero.
- A Canada total adds up only the places that report: hospital stays leave out Quebec and emergency department visits cover 7 of 13 provinces and territories.
- Quarters add up to as much as 1.57% less than the published year, so I never build a year from quarters.
- 2026 covers January to March only. Some 2025 values look provisional (Ontario deaths -37.49%, Quebec +93.14%), so I compare provinces on 2024.
- Checks: no duplicate keys, Canada totals reconciled to the places that report, and every headline number checked against a separate Python calculation.

Terms in plain language

- Crude rate: cases per 100,000 people living in that place, not adjusted for age.
- Suppressed: the agency hides small counts to protect privacy, so the true value is unknown, not zero.
- Not available: the source does not publish this value for that place or period.
- Hospital stay: a patient admitted to hospital. Emergency department visit: seen in the emergency room. EMS response: an ambulance call.
- Index (2019 = 100): each value divided by that source's 2019 value, so sources of different size can be compared.
- Provisional: the newest figures can be revised, so treat the latest year with care.

Data: PHAC, Opioid- and Stimulant-related Harms in Canada (Open Government Licence - Canada). No individual records.

Counts to 2026 Q1 | Latest full year 2025 | 2026 is January to March only